

Drop-in text for Sec. VI-E (deviation bound) and Sec. VI-B (ramp tracking)

numbering continues from Eq. (86); symbols follow the manuscript

VI-E continuation: impulse response and the bound on e_φ

After Proposition 1 / Eq. (86).

The homogeneous part of (86) is spanned by $\cosh C_2\tau$ and $\sinh C_2\tau$ with $C_2 = \sqrt{Wz_{CoM}/J_P}$ from (20). Its impulse response — the solution of $J_P\ddot{h} - Wz_{CoM}h = \delta(\tau)$ from rest — follows from the momentum jump $J_P[\dot{h}(0^+) - \dot{h}(0^-)] = 1$, the position being unable to jump:

$$h_\varphi(\tau) = \frac{\sinh(C_2\tau)}{J_PC_2}, \quad h_\varphi(0) = 0, \quad \dot{h}_\varphi(0) = \frac{1}{J_P}. \quad (87)$$

Because $h_\varphi(0) = 0$, the boundary term generated by differentiating a convolution with a variable upper limit,

$$\frac{d}{d\tau} \int_0^\tau h_\varphi(\tau-s)\rho(s)ds = \underbrace{h_\varphi(0)}_{=0} \rho(\tau) + \int_0^\tau \dot{h}_\varphi(\tau-s)\rho(s)ds,$$

vanishes for *any* forcing. The zero initial conditions in (86) therefore set both homogeneous coefficients to zero — no assumption on $\rho(0)$ is required — and the deviation is the Duhamel integral alone,

$$e_\varphi(\tau) = \int_0^\tau h_\varphi(\tau-s)\rho(s)ds. \quad (88)$$

The two channels. Two terms of (79) leave the estimator span: the gravity remainder (83) and the bilinear ground-effect coupling (79)(iv),

$$\rho(\tau) = \rho_\varphi(\tau) + \rho_{GE}(\tau), \quad \rho_{GE}(\tau) = \beta_M \dot{M}\tau\varphi. \quad (89)$$

Both vanish at the onset — with $\varphi_{\text{nom}} \sim \tau^3$ the gravity remainder is $\mathcal{O}(\varphi^2) \sim \tau^6$ and the bilinear term is $\tau \cdot \tau^3 = \tau^4$ — so each is largest at the window end.

A bound that can be verified by hand. Bounding each channel there and integrating the kernel, $\int_0^\tau h_\varphi(u)du = (\cosh C_2\tau - 1)/(J_PC_2^2)$, and using $J_PC_2^2 = Wz_{CoM}$,

$$|e_\varphi(\tau_{\text{end}})| \leq \frac{\rho_{\varphi,\text{max}} + \rho_{GE,\text{max}}}{Wz_{CoM}} (\cosh x - 1), \quad x \triangleq C_2\tau_{\text{end}}. \quad (90)$$

The inertia cancels: (90) requires only Wz_{CoM} , the dimensionless group x and the two channel maxima, all of which are already reported.

Each maximum is elementary. The gravity remainder is $\rho_\varphi = \frac{1}{2}G''(\varphi^*)\varphi^2$ with $G'' = W(l_p + \lambda_{\text{off}})\cos\varphi^* - Wz_{CoM}\sin\varphi^*$ from (84), so its worst case over the operating box is reached at the tilt cap, $\rho_{\varphi,\text{max}} \approx 71 \text{ mNm}$ at $\varphi_{\text{max}} = 10^\circ$; over the runs actually flown it is 2.7 mNm in the median and 12.3 at worst. The coupling is a product of two measured quantities, $\rho_{GE,\text{max}} = |\beta_M|\Delta M_{\text{win}}\varphi_{\text{end}}$ with $\Delta M_{\text{win}} = \dot{M}\tau_{\text{end}}$ the moment increment over the window; with $\beta_M = -0.0345 \text{ rad}^{-1}$ from Table 2 that is 3.5 mNm at the box corner ($\Delta M_{\text{win}} = 0.59 \text{ Nm}$, $\varphi = 10^\circ$) and 0.7 mNm in the median over the flown runs. The gravity channel therefore dominates by a factor of about twenty.

Nothing beyond $|\rho| \leq \rho_{\text{max}}$ enters (90): no monotonicity, no shape, and no property of the trajectory but the window length. That is deliberate. The bound is the hand-checkable companion to the per-run propagation, which convolves the measured ρ with the exact kernel and needs no envelope at all; the entry reported in Table ?? is the latter, and (90) exists so that its order of magnitude can be confirmed from Wz_{CoM} , x and the channel maxima alone.¹

¹The bound tightens if ρ is additionally taken monotone increasing, which it is here. Chebyshev's integral inequality for

Relative form: both J_P and Wz_{CoM} cancel. Dividing (90) by the nominal excursion $\varphi_{\text{nom}}(\tau) = (C_1/C_2)(\sinh C_2\tau - C_2\tau)$ with $C_1 = M/Wz_{CoM}$, and writing $\Delta M_{\text{win}} = \dot{M}\tau_{\text{end}}$ for the moment increment over the fit window,

$$\frac{|e_\varphi|_{\text{end}}}{\varphi_{\text{nom},\text{end}}} \leq \frac{\Psi(x)\bar{\rho}}{\Delta M_{\text{win}}}, \quad \Psi(x) = \frac{x(\cosh x - 1)}{\sinh x - x}. \quad (91)$$

Each product is bounded uniformly in x — ΨR_φ rises from $3/7$ to $1/2$ and ΨR_{GE} from $2/5$ to 1 — so

$$\boxed{\frac{|e_\varphi|_{\text{end}}}{\varphi_{\text{nom},\text{end}}} \leq \frac{\frac{1}{2}\rho_{\varphi,\text{max}} + \rho_{b,\text{max}}}{\Delta M_{\text{win}}}, \quad |\Delta M_{\text{crit}}| \leq \frac{\rho_{\varphi,\text{max}}}{7} + \frac{\rho_{b,\text{max}}}{5}} \quad (92)$$

independently of C_2 , J_P and Wz_{CoM} . The second inequality follows because the onset minimises the two-segment residual: linearising its stationarity condition gives $\Delta M_{\text{crit}} = -\int_0^{\tau_{\text{end}}} \rho w$ with a non-negative weight normalised to $\int w = 1$, so $|\Delta M_{\text{crit}}| \leq \bar{\rho}$. A reviewer can evaluate (93) from numbers already in the paper.

Evaluation. Every quantity must be read off the *same* window: $\rho_{\varphi,\text{max}}$ and $\rho_{b,\text{max}}$ at the excursion that window reaches, ΔM_{win} as its own moment increment. Over the 140 gate-passing runs, with $W = 31.59$ N, $l_p + \lambda_{\text{off}} = 0.160/0.130$ m (roll/pitch), $z_{CoM} = 0.2$ m (conservative: the CAD value is 0.256 m and ρ_φ decreases with z_{CoM} , by 2% at the largest tilt) and $\beta_M = 0.0345$ rad $^{-1}$:

$\dot{M}$ [N m/s]	0.10	0.30	0.65	1.20	median	worst
$ \varphi _{\text{end}}$ [deg]	5.23	5.90	6.55	6.17	5.95	9.33
ΔM_{win} [N m]	0.072	0.165	0.306	0.479	0.221	—
$\rho_{\varphi,\text{max}}$ [mN m]	19.7	22.5	30.4	27.2	23.6	58.3
$\rho_{b,\text{max}}$ [mN m]	0.24	0.59	1.24	1.85	0.68	3.00
$x = C_2\tau_{\text{end}}$	3.47	2.64	2.15	1.79	2.67	5.20
$ e_\varphi /\varphi_{\text{nom}}$ [%]	13.8	6.0	4.7	2.9	4.9	25.1
$ e_\varphi $ [deg]	0.76	0.34	0.31	0.18	0.32	1.68
$ \Delta M_{\text{crit}} $ [mN m]	1.94	2.73	3.77	3.76	3.01	7.86
$ \Delta\lambda_{\text{off}} $ [mm]	0.061	0.086	0.119	0.119	0.095	0.249
ground-effect share of $\bar{\rho}$ [%]	1.8	4.0	6.0	8.8	5.1	25.6

Using instead the x -free constants of (93) the worst case becomes $|\Delta M_{\text{crit}}| \leq 8.8$ mN m, i.e. $|\Delta\lambda_{\text{off}}| \leq 0.28$ mm — the price of not evaluating R_φ, R_{GE} is under 12%.

The bilinear ground-effect channel contributes 5.1% of $\bar{\rho}$ in the median and 25.6% at worst; it grows with the ramp rate because $\rho_{GE} \propto \dot{M}\tau$, whereas the gravity remainder depends on the excursion alone. [The threshold bound \$|\Delta\lambda_{\text{off}}| \leq 0.25\$ mm is the figure that matters: it does not involve \$\Delta M_{\text{win}}\$, and so does not inherit the loss the relative normalisation suffers at slow ramps.](#) The relative bound is loosest at 0.10 N m/s, where the window closes on a small moment increment while the excursion is unchanged — the same ordering the excitation design reports — and it is conservative by roughly a factor 5 overall, since it forwards the *unprojected* remainder (23.6–58.3 mN m here, against 1.2–12.3 mN m measured after the estimator absorbs its own degrees of freedom).

VI-B: ramp input tracking (rate limits)

The rotor response identified in Sec. V (Table 4: $\omega_{n,\text{rot}} = 21.74$ rad/s, $\zeta = 0.7814$, $\tau_d = 15.67$ ms) tracks a ramp with the steady-state lag

$$t_{\text{lag}} = \frac{2\zeta}{\omega_{n,\text{rot}}} + \tau_d = 71.9 + 15.7 = 87.6 \text{ ms},$$

so the realised moment trails the command by $\dot{M} t_{\text{lag}}$, i.e. 8.8 mN m at 0.10 N m/s and 105 mN m at 1.20 N m/s.

functions of *opposite* monotonicity — ρ rises from zero at the onset while the kernel $\cosh(C_2(\tau - s))$ decreases in s — replaces each maximum in (90) by its time average. For the nominal shapes $\rho_\varphi \propto (\sinh u - u)^2$ and $\rho_{GE} \propto u(\sinh u - u)$ with $u = C_2\tau$ those averages are $R_\varphi(x)\rho_{\varphi,\text{max}}$ and $R_{GE}(x)\rho_{GE,\text{max}}$ with $R_\varphi \leq 1/7$ and $R_{GE} \leq 1/5$, the limits being the $x \rightarrow 0$ values where the two channels are sixth and fourth order; over $x \in [1.08, 5.20]$ they are 0.089–0.139 and 0.145–0.197, worth a factor of 5 to 11. We quote (90) without this refinement, since the conclusion does not need it and the refinement costs an assumption on ρ that the envelope does not.

This lag does *not* bias M_{crit} . The identification consumes the *realised* moment, reconstructed from the measured rotor speeds, not the commanded one; a common lag therefore shifts the moment trace and the rate trace together and leaves the moment read at the detected onset unchanged. What the lag does consume is window length, which the sample-count gate enforces.

Two residual effects remain and are both gated. First, the realised *slope* may differ from the commanded rate; the run-level gate rejects $|\dot{M}_{\text{meas}} - \dot{M}_{\text{cmd}}|/\dot{M}_{\text{cmd}} > 3\%$, and since $C_1 = K\dot{M}$, a 3% rate error perturbs the amplitude constant by 3% — well inside the $\pm 20\%$ mis-specification the sensitivity analysis shows to be harmless. Second, the start of the ramp is not linear while the rotor transient decays; with $4/(\zeta\omega_{n,\text{rot}}) = 235$ ms to 2%, that transient is confined to the low-moment part of the window, which the slope gate excludes by scoring only the segment above the per-axis moment floor. The linearity gate (30 mN m RMSE about the best-fit line) then acts as a fault detector for ramps that are stepped or aborted outright.